

MATH3881/5881: Statistical Machine Learning Theory

Week 4: Theory Tutorial Solutions: Backpropagating the Variance

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Worked solutions to the Week 4 Theory Tutorial. Numbering matches the problem sheet, and notation follows the Background section there: L -layer width- n ReLU network, weight entries $w_{ij}^{[\ell]}$ iid with mean 0 and variance s_W^2 , gating matrix $D^{[\ell]} = \text{diag}(\mathbf{1}[z^{[\ell]} > 0])$, and $V_\ell := \mathbb{E}[\|\delta^{[\ell]}\|^2]$.

Problem 1: The variance recursion

Step 1. The backprop recursion $\delta^{[\ell]} = D^{[\ell]}(W^{[\ell+1]})^\top \delta^{[\ell+1]}$ gives, component-wise,

$$\delta_j^{[\ell]} = D_{jj}^{[\ell]} \sum_{i=1}^n w_{ij}^{[\ell+1]} \delta_i^{[\ell+1]}.$$

Step 2. Square and sum over j . Use $(D_{jj}^{[\ell]})^2 = D_{jj}^{[\ell]}$, which holds because $D_{jj}^{[\ell]} \in \{0, 1\}$:

$$\|\delta^{[\ell]}\|^2 = \sum_{j=1}^n D_{jj}^{[\ell]} \left(\sum_{i=1}^n w_{ij}^{[\ell+1]} \delta_i^{[\ell+1]} \right)^2.$$

Step 3. Expand the inner square as a double sum,

$$\left(\sum_{i=1}^n w_{ij}^{[\ell+1]} \delta_i^{[\ell+1]} \right)^2 = \sum_{i=1}^n \sum_{k=1}^n w_{ij}^{[\ell+1]} w_{kj}^{[\ell+1]} \delta_i^{[\ell+1]} \delta_k^{[\ell+1]}.$$

Step 4. Condition on $\delta^{[\ell+1]}$, and apply the Background's working assumption that $W^{[\ell+1]}, D^{[\ell]}, \delta^{[\ell+1]}$ are mutually independent. Take the expectation over $W^{[\ell+1]}$ first, with $\delta^{[\ell+1]}$ and $D^{[\ell]}$ held fixed. The vanishing cross-terms fact gives $\mathbb{E}[w_{ij}^{[\ell+1]} w_{kj}^{[\ell+1]}] = s_W^2 \mathbf{1}[i = k]$, so the double sum collapses:

$$\mathbb{E} \left[\sum_{i,k} w_{ij}^{[\ell+1]} w_{kj}^{[\ell+1]} \delta_i^{[\ell+1]} \delta_k^{[\ell+1]} \mid \delta^{[\ell+1]}, D^{[\ell]} \right] = s_W^2 \sum_{i=1}^n (\delta_i^{[\ell+1]})^2 = s_W^2 \|\delta^{[\ell+1]}\|^2.$$

Step 5. Substituting back into Step 2,

$$\mathbb{E}[\|\delta^{[\ell]}\|^2 \mid \delta^{[\ell+1]}, D^{[\ell]}] = s_W^2 \|\delta^{[\ell+1]}\|^2 \sum_{j=1}^n D_{jj}^{[\ell]}.$$

Step 6. Take the expectation over $D^{[\ell]}$, with $\delta^{[\ell+1]}$ still held fixed. The active-neuron rate at initialisation gives $\mathbb{E}[D_{jj}^{[\ell]}] = 1/2$, hence

$$\mathbb{E}\left[\sum_{j=1}^n D_{jj}^{[\ell]}\right] = \sum_{j=1}^n \mathbb{E}[D_{jj}^{[\ell]}] = \frac{n}{2}.$$

So

$$\mathbb{E}[\|\delta^{[\ell]}\|^2 \mid \delta^{[\ell+1]}] = s_W^2 \|\delta^{[\ell+1]}\|^2 \cdot \frac{n}{2} = \frac{n s_W^2}{2} \|\delta^{[\ell+1]}\|^2.$$

Step 7. Take the outer expectation over $\delta^{[\ell+1]}$:

$$V_\ell = \mathbb{E}[\|\delta^{[\ell]}\|^2] = \frac{n s_W^2}{2} \mathbb{E}[\|\delta^{[\ell+1]}\|^2] = \frac{n s_W^2}{2} V_{\ell+1}.$$

This is the variance recursion.

Step 8. Iterating the recursion $L - 1$ times, starting from $V_L = \|\delta^{[L]}\|^2$ (deterministic, since $\delta^{[L]} = a^{[L]} - y$ was fixed),

$$V_1 = \left(\frac{n s_W^2}{2}\right)^{L-1} V_L = \left(\frac{n s_W^2}{2}\right)^{L-1} \|\delta^{[L]}\|^2.$$

Problem 2: The unique stable scaling

Step 1. From Problem 1,

$$\frac{V_1}{V_L} = \left(\frac{n s_W^2}{2}\right)^{L-1}.$$

Step 2. Write $r := n s_W^2/2$, so $V_1/V_L = r^{L-1}$. Analyse the three cases as $L \rightarrow \infty$:

- If $r > 1$: $r^{L-1} \rightarrow \infty$, so $V_1/V_L \rightarrow \infty$ (exploding gradient at layer 1).
- If $r < 1$: $r^{L-1} \rightarrow 0$, so $V_1/V_L \rightarrow 0$ (vanishing gradient at layer 1).
- If $r = 1$: $r^{L-1} = 1$ for every L , so $V_1/V_L = 1$ (stable gradient).

Step 3. Only the third case avoids both pathologies, so the requirement forces $r = 1$, i.e., $n s_W^2/2 = 1$, which rearranges to

$$s_W^2 = \frac{2}{n}.$$

This is the He criterion for ReLU networks.

Step 4. Interpretation: the factor of 2 is the inverse of the active-neuron rate $1/2$. ReLU silences half the gradient at each backward layer, so to keep V_ℓ stable the weight variance must double, from the linear/Xavier value $s_W^2 = 1/n$ (where $\mathbb{E}[D_{jj}^{[\ell]}] = 1$ would replace $1/2$) to $s_W^2 = 2/n$. The factor of 2 is exactly the compensation for ReLU's gating.